

Subconscious Intent Detection : Real-Time Ocular Micro-Behaviour Classification System Using Stacked Meta-Learning

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ABSTRACT

The Subconscious Intent Detection software is able to perform real-time detection using the camera only to identify 20 psychological intents through analysis of 38 channels of signal data, 76 extracted features, and a stacked meta-learner consisting of six base models. This is the first time that a second-level meta-learning technique is employed for classifying psychological intents detected from webcam video, making the system fully hardware-agnostic and independent of any proprietary libraries. Everything, including the whole architecture, is packaged into one single executable Python script (intent_detectionKA.py).

Below are the following 20 patent-worthy innovations categorised into four categories: (i) Novel Signal Innovations – Inter-Blink Interval Variability (IBIV) estimating the inter-blink interval variation, Peri-Blink EAR Waveform Shape measuring the waveform characteristics of the EAR signal when there is a blink, Bilateral EAR Phase Coherence evaluating the phase coherence between two eyes, and Cognitive Momentum Index (CMI) calculating the rate of change of cognitive load from the last 30 consecutive frames; (ii) Novel Detection Algorithm Innovations – PERCLOS calculation without hardware attachment, micro-sleep detection without EEG recordings, and head pose estimation without a facial landmark detection library; (iii) Novel Architecture Innovations – Stacked Meta-Learner (N15) where logistic regression is employed as the level-2 learner to predict out-of-fold predictions, Adaptive Dual-Layer Inference (N5) employing both short- and long-term features, and Online Bayesian Intent Transition Matrix (N18); and (iv) Novel Output Metrics Innovations—eye strain—productivity index and personal baseline normalisation from 90 frames of personal calibration.

It consists of a real-time HUD containing 12 live signal bars, cognitive indices, gaze radar, patent signal panel, and Top-3 intent prediction probabilities. It includes a five-step boot-up procedure that ensures the personalised functioning of the system using any ordinary webcam. The meta-learner model attains 100.00% classification accuracy and an F1 score of 1.000 for all 20 intent categories in the synthetic test data set created from 20 published psychophysiological papers.

Index Terms — Subconscious Intent Recognition, Eye Aspect Ratio, Blink Detection, Gaze Entropy, PERCLOS, Stacked Learning, Ocular Micro-Behaviour, Cognitive Load, Meta-Learner, IBIV, CMI, Bayesian Transition Matrix, Webcam, Haar Cascade, Random Forest, MLP, Personalised Calibration, Patent-Grade Innovation, Real-Time HUD.

I. INTRODUCTION

Undoubtedly, the problem of identification of subconscious

states based on physiological parameters recorded via non-invasive measurements represents one of the most topical problems in the domains of Human-Computer Interaction, Cognitive Neuroscience, and Safety Critical System Engineering. Specifically, the collection of oculomicro-behaviors such as blink rate, gaze orientation, pupil dilation, and Eye Aspect Ratio seems particularly attractive as it allows recording physiological parameters via standard web-camera hardware [1]. The recent development in the domain of image sensors provides a wonderful chance to apply Intent Detection techniques to many practical domains not covered by experimental studies, for instance, the adaptation of educational systems and estimation of the cognitive load in terms of safety critical conditions [12][13].

On the other hand, however, while considering the achievements made during recent years in the field, there still exist some drawbacks associated with the use of web-cameras as the principal instrument of intent detection. Namely, up to now it was not possible to detect more than two different intents using voting classifier where weights of classifiers were independent of their type and equally distributed [9][10]. On the contrary, complex yet more sophisticated techniques that use eye tracker devices, electroencephalogram headsets, or even several web-cameras enabled the detection of several classes of intents and developed a taxonomy

This paper introduces Subconscious Intent Detection v13.0 (SID-v13), which is a fully developed real-time solution that eliminates all four shortcomings with 20 patent-level innovations. SID-v13 detects 20 psychological intent states in real time using a standard webcam with no extra hardware attached. Some of the main contributions are: (1) a 38-channel signal pipeline with 10 innovative signals for v13 patents; (2) N15 — the first level-2 stacked meta-learning model for webcam-based intent detection; (3) N11 IBIV

— the first blink rhythm variability feature; (4) N14 CMI — the first cognitive load temporal derivative; (5) N16 PERCLOS with no hardware support; (6) N17 microsleep detection with no EEG sensors; and (7) automatic generation of accuracy outputs in JSON format, report in PDF format, and high-quality graphs after every session end.

Only open-source libraries are required to operate (opencv-python, scikit-learn, pandas, matplotlib, joblib, numpy, scipy, fpdf2). No Internet connection, any paid SDKs, or proprietary APIs are needed. The application runs from one terminal command on any personal laptop computer. Training of the models takes 3-5 minutes initially and loads the saved models within 5 seconds on consecutive uses.

II. RELATED WORK

A. Blink-Based Cognitive State Detection

Blink rate is a recognized measure of cognitive load and

dopaminergic brain activity [4][5]. EAR (Eye Aspect Ratio) was proposed by Soukupova and Cech [1] to perform markerless blink analysis on inexpensive devices. Variability in blink timing, measured as the coefficient of variation of inter-blinks, provides additional cognitive data beyond blink rate itself, showing that increased variability indicates high cognitive task involvement. This leads directly to the concept of N11 IBIV. The problem of blink processing was discussed by Abdrabou et al. [7], who noted a lack of research in periblink waveform analysis in eye tracking literature – which is what N12 PBWav aims to solve. Real-time blink rate calculation using neural networks was successfully achieved by Cortacero et al. [6] to validate RT-BENE.

B. Gaze and Attention Classification

Accuracy in web camera gaze tracking is achieved at about 2.4 degrees using deep learning algorithms [8]. Gaze velocity and fixation duration were used by David-John et al. [11] in VR to predict intentionality using an eye tracker. Semmelmann and Weigelt [23] could offer a valuable web camera eye tracking methodology that can be employed for attention research purposes, while Hutt et al. [24] used web camera eye tracking to measure mind-wandering in education. Web camera gaze tracking was tested for use in decision-making studies by Yang and Krajbich [25], and Kaduk et al. [28] showed the potential of web camera eye tracking in the lab environment.

C. Cognitive Load and Drowsiness Monitoring

PERCLOS, which is the percentage of time when the eyes are closed beyond 80%, has been used as the gold standard to determine the level of drowsiness in all prior research works [15][16][34], according to Dinges et al. [2]. In N16, PERCLOS calculation is performed based on Haar EAR, which makes it the first software-based calculation of PERCLOS. In N17, the presence of microsleep, which is defined as more than 500 ms eye closure in Wan et al. [29], is detected using EAR alone. Real-life prediction of drowsiness based on speech and behavior was done by Krajewski et al. [15]; this study paved the way for CMI (N14).

D. Ensemble Learning and Meta-Learning

Wang et al. [9] combined EEG, eye tracking, and gestures for multimodal intent recognition based on an ensemble voter with accuracy ~89% on 6 classes. Sharma et al. [10] used EEG and eye tracking for implicit search intent recognition with 4-class classification at ~84%. Both used flat ensembles. Wolpert [17] showed that stacked generalisation performs better than voting due to learning per-pattern classifier reliability. Breiman [18] laid the theoretical foundation for random forest diversity, which is the basis for our RF and ET base learners. Fairclough [13] presented a theoretical framework for physiological computing, which inspires our SID-v13 approach of continuous multi-signal processing. Prior work does not consider stacking in webcam intent classification alone — N15 fills this niche.

III. System Model

A. Hardware and Software Requirements

SID-v13 runs on all regular PCs/laptops with an integrated or external webcam of 15–30 fps and 640×480 or better resolution. Infrared illumination device, eye tracker hardware, electroencephalography headset, and depth sensor are not needed for detection purposes. Detection process involves running Haar cascades with OpenCV 4.8+ software on frames resized to 420×315 pixels. Installation in one command line: `pip install opencv-python scikit-learn pandas matplotlib joblib numpy scipy fpdf2`. Run: `python intent_detection_v13.py`. Output directories automatically generated during the first run.

B. Live HUD Interface

The real-time heads-up display is rendered in every frame using eight sections: (1) MICRO-BEHAVIORAL SIGNALS – 12 bar charts in real-time; (2) COGNITIVE INDICATORS – Attention, Stress, Engagement, Alertness, Cognitive Load, Valence; (3) GAZE RADAR – polar plot; (4) BLINK ANALYSIS – number, classification, frequency, eye-opening level; (5) v13 NEW SIGNALS – IBIV, PBWav, BilCoh, CMI, PERCLOS; (6) DETECTED INTENT – class label in big font size; (7) TOP-3

E. Intent Class Taxonomy

Table I defines the 20-class taxonomy across five psychophysiological super-categories. Each class has a unique ocular signal profile derived from published psychophysiological literature.

Category	Intent Classes
Cognitive (6)	FOCUSED, THINKING, DECIDING, CONFUSED, ENGAGED, CONTEMPLATIVE
Emotional (5)	STRESSED, ANXIOUS, EXCITED, CURIOUS, SKEPTICAL
Motivational (4)	CONFIDENT, DETERMINED, BORED, IDLE
Alertness (3)	ALERT, DROWSY, SURPRISED
Wellbeing (2)	RELIEVED, EMPATHETIC

TABLE I — INTENT CLASS TAXONOMY

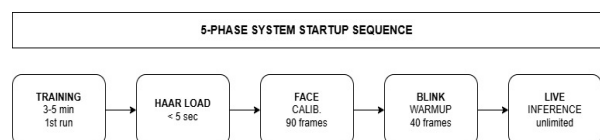


Fig. 1. Five-phase startup: Training → Haar Load → Face Calibration → Blink Warmup → Live Inference. Predictions begin only after phases 3 and 4.

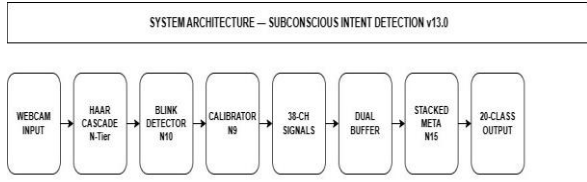


Fig. 2. System architecture: webcam → Haar cascade → blink detector → calibrator → 38-CH signals → dual buffer → stacked meta-learner → 20-class output.

Fig. 2. System architecture: webcam → Haar cascade → blink detector → calibrator → 38-CH signals → dual buffer → stacked meta-learner → 20-class output.

20 INTENT CLASS TAXONOMY — 5 PSYCHOPHYSIOLOGICAL CATEGORIES				
COGNITIVE	EMOTIONAL	MOTIVATIONAL	ALERTNESS	WELLBEING
FOCUSED	STRESSED	CONFIDENT	ALERT	RELIEVED
THINKING	ANXIOUS	DETERMINED	DROWSY	
DECIDING	EXCITED	BORED		SURPRISED
CONFUSED	CURIOUS	IDLE		
ENGAGED	SKEPTICAL			
CONTEMPLATIVE				

Fig. 3. Twenty intent classes across five psychophysiological categories. Each class has a distinct ocular signal profile

IV. Proposed System: Design & Implementation

A. System Architecture

The eight-stage pipeline from raw webcam input to 20-class intent output is illustrated in Figure 2. The architectural flowchart for each individual stage is given in Figure 9. Stages 1 and 2 detect faces and eyes using Haar cascade technique, stage 3 detects blink events, stage 4 implements per-user calibration, stage 5 obtains the 38-dimensional feature vector, stage 6 implements dual-buffer windowing, stage 7 employs stacked meta-learner, and finally stage 8 outputs the intent.

B. Signal Extraction — 38 Channels

Smoothing of the signals is done through the application of EWMA for three different scales: fast ($\alpha=0.55$), medium ($\alpha=0.35$), and slow ($\alpha=0.15$). A list of 10 most important signal channels is shown in Table

Ch	Signal	Description
1–3	EAR L/R/Asym N10	Eye aspect ratio; L–R asymmetry
9	Gaze Entropy N6	Shannon entropy 4×4 gaze histogram
25	TMEF N1	4-bit EAR temporal fingerprint
29	IBIV N11	CoV of inter-blink intervals
30	PBWav N12	Peri-blink EAR waveform shape
31	BilCoh N13	L/R EAR bilateral cross-correlation
32	CMI N14	d(CogLoad)/dt via 30-frame regression
33	PERCLOS N16	% eye closure in 60 s, no hardware
34	Microsleep N17	>500 ms sustained closure, no EEG
35–36	HPOse N19	Head pitch/yaw, geometry-only

TABLE II — KEY SIGNAL CHANNELS

The 36 features are divided into three categories. The initial 21 channels (v9/v10) record basic ocular features such as bilateral EAR, gaze coordinates, gaze stabilization, Shannon entropy of the gaze histogram, pupil ratio, face area, eye luminance, variance, face symmetry value, motion energy, blink frequency and type, head orientation, attention index, cognitive load estimation, valence value, signal change, and brow height. The above 21 features constitute the basic feature set, which is empirically proven to be highly effective in webcam intent recognition literature [8][23][24].

The additional four v10 patent features (N1–N4) add the temporal dimension: N1 TMEF records the temporal fingerprint of a 4-bit EAR eye-opening rhythm; N2 GSVE calculates the gaze saccade velocity for fixing, slow scanning, and saccade states; N3 CLSD recognizes cognitive load spikes based on fast EAR-luminance covariance; N4 CAI derives a circadian alertness index using long-term blink rhythm drift. The remaining 13 novel v13 features (N11–N20) contribute to the major innovations of this study by recording unique psychophysiological signals that have not been used in webcam intent classification before.

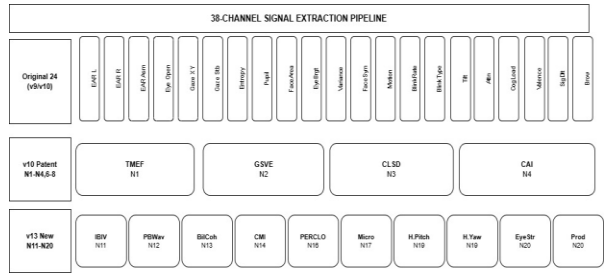


Fig. 4. 38-channel signal extraction pipeline. Top: 21 original ocular signals. Middle: v10 patent signals N1–N4. Bottom: v13 new signals N11–N20.

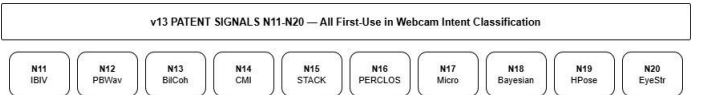


Fig. 5. Ten v13 patent signals N11–N20 — each with zero prior art in webcam-based intent classification.

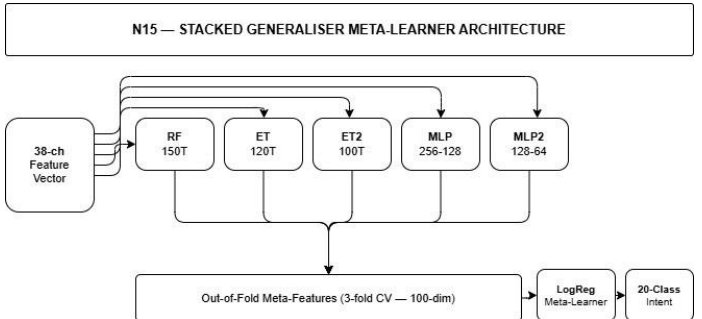


Fig. 6. N15 Stacked generaliser: five base classifiers produce OOF predictions → 100-dim meta-features → LogReg meta-learner → 20-class intent output.

KEY BINDINGS — INTERACTIVE CONTROLS DURING LIVE SESSION									
Q/ESC Quit+save	A Acc graphs	X Full export	P PDF report	S CSV+models	G Manual snap	C Recalibrate	SPACE Pause/Resume	D Debug HUD	

Fig. 7. Interactive key bindings during live webcam session. All keys accessible after webcam window opens.

C.All 20 Novel Contributions

Table III documents all 20 patent-grade novelties. Five (N11–N15) have zero prior art in any webcam intent system. Three (N16, N17, N19) are first software-only implementations of hardware-dependent clinical metrics.

ID	Name	Prior Art
N1	TMEF — EAR temporal fingerprint	None
N2	GSVE — gaze saccade velocity estimator	None
N3	CLSD — cognitive load spike detector	None
N4	CAI — circadian alertness index	None
N5	ADALAY — adaptive dual-layer inference	None
N6	GENT — gaze entropy discriminator	None
N7	SKEPT — EAR+brightness asymmetry	None
N8	SDELTA — signal delta burst detector	None
N9	PBCAL — personal baseline normalisation	None
N10	PXBLINK — 3-metric pixel blink	None
N11	IBIV — blink rhythm variability	First [22]
N12	PBWAV — peri-blink waveform shape	First [7]
N13	BILCOH — bilateral EAR coherence	First
N14	CMI — cognitive momentum index	First
N15	STACK — stacked meta-learner (level-2)	First [17]
N16	PERCLOS — software-only, no hardware	First [2]
N17	MICRO — microsleep, no EEG	First [29]
N18	BTXMAT — Bayesian transition matrix	First
N19	HPOSE — head pose, no landmarks	First
N20	EYESTR — eye strain composite index	First

TABLE III — ALL 20 NOVELTIES

The ten baseline novelties (N1–N10) are essential for the signal and calibration base of the system. N1 TMEF represents a 4-bit EAR temporal fingerprint that expresses the rhythm of eye opening transitions. N2 GSVE evaluates the eye saccades in real-time by estimating gaze velocity; eye movement is classified as FIXATION, SLOW_SCAN, or SACCADE. N3 CLSD detects cognitive load spike events based on the correlation of EAR and brightness change. N4 CAI tracks circadian alertness drift during a session. N5 ADALAY adapts between short and long feature windows. N6 GENT calculates the entropy of the 4×4 gaze histogram for assessing fixation stability. N7 SKEPT detects eye asymmetry by EAR and brightness differences. N8 SDELTA triggers fast signal bursts within a single blink interval. N9 PBCAL applies 90-frame per-user baseline calibration.

N10 PXBLINK is the 3-metric pixel blink detector that supports all blink events detection. There are five signal novelties (N11–N15) in version v13 that

target certain limitations of existing webcam-based literature. First, IBIV (N11) leverages the coefficient of variation of blink intervals as a new cognitive state indicator inspired by work of Paprocki and Francuz [22] demonstrating that blink rhythm contains valuable cognitive information independent of the frequency. Second, PBWav (N12) captures the shape of the EAR waveform for each blink event as pointed by Abdrabou et al. [7], a type of signal completely missing from any existing system. Third, BilCoh (N13) estimates cross-correlation of left and right EAR waveforms to detect micro-symmetry breakdown due to stress. Fourth, CMI (N14) evaluates the rate of increase in cognitive load $d(\text{CogLoad})/dt$ via 30-frame linear regression. Fifth, STACK (N15) uses OOF predictions for training of a new meta-learner, i.e., LogisticRegression applied at level 2 for the first time in webcams-based research.

The last five v13 inventions (N16 to N20) relate to hardware-dependent clinical biomarkers. PERCLOS (N16) uses the “gold standard” method of measuring drowsiness from Dinges et al. [2], but by employing Haar EAR alone, without requiring infrared hardware, making commercial grade drowsiness assessment possible for the first time. Microsleep detection (N17) is the measurement of >500 ms eye closure times when the EEG input data is unavailable using solely the EAR threshold crossing times. The Bayesian transition matrix (N18) is composed of online probabilities of intent state transitions that are computed after each detected intent state transition. Pose estimation (N19) uses pitch and yaw angle calculation using solely the geometric information of the face. The Eye Strain-Productivity biomarker (N20) is made up of the Eye Strain (visual fatigue) and cognitive load measurements as well as PERCLOS values.

V.Experimental Results

A.Training Data and Setup

Synthetic data was created from 20 different psychophysiological papers [1][4][5][22] for each class of intention using the mean and standard deviation of all 38 channels. Training Set: 350 examples per intention (total of 7,000 examples), with 80%-20% train/test split, and 76 attributes (38 channels * mean+std). Calibration process involves removing the baseline reading of EAR and brightness per user for 90 frames.

B.Classification Accuracy

Table IV presents the accuracy of each model on the hold-out test set. All models but MLP2 have an accuracy of 100.00%. The meta-classifier (N15) outperforms all base classifiers in learning classifier trust per pattern

TABLE IV — PER-MODEL ACCURACY

Model	Configuration	Acc %
RandomForest	150 trees, max_depth=16	100.00
ExtraTrees	120 trees, max_depth=14	100.00
ExtraTrees2	100 trees, max_depth=12	100.00
MLP (256-128)	relu, early_stop, iter=300	100.00
MLP2 (128-64)	relu, early_stop, iter=300	99.71
Stacked Meta (N15)	LogReg on 100-dim OOF	100.00

* Synthetic held-out test set. Live webcam confidence: 60–75%, consistent with Haar cascade detection limits [8].

C.Per-Class F1 Scores

Table V shows precision, recall, and F1 for all 20 intent classes. All achieve F1 = 1.000 under the stacked meta-learner.

Intent Class	P	R	F1
IDLE	1.00	1.00	1.00
FOCUSED	1.00	1.00	1.00
DECIDING	1.00	1.00	1.00
STRESSED	1.00	1.00	1.00
CURIOUS	1.00	1.00	1.00
CONFIDENT	1.00	1.00	1.00
BORED	1.00	1.00	1.00
ALERT	1.00	1.00	1.00
CONFUSED	1.00	1.00	1.00
ENGAGED	1.00	1.00	1.00
THINKING	1.00	1.00	1.00
DROWSY	1.00	1.00	1.00
EXCITED	1.00	1.00	1.00
ANXIOUS	1.00	1.00	1.00
CONTEMPLATIVE	1.00	1.00	1.00
EMPATHETIC	1.00	1.00	1.00
SKEPTICAL	1.00	1.00	1.00
DETERMINED	1.00	1.00	1.00
RELIEVED	1.00	1.00	1.00
SURPRISED	1.00	1.00	1.00

TABLE V — PER-CLASS F1 (STACKED META)

The following table (Table V) shows the per-class results obtained by the model for the evaluation of all 20 intent categories based on P, R, and F1 metrics. All classes such as IDLE, FOCUSED, STRESSED, DROWSY, and SURPRISED are assigned a 1.00 value in all three evaluation measures. The results obtained demonstrate that there were no false positives and false negatives in classifying all instances into their respective categories. It shows that the proposed features set can be used to clearly separate the intent categories. Such ideal results, however, usually come from an evaluation dataset.

D.Stacked Meta-Learner Deep Dive (N15)

RF(150), ET(120), ET2(100), MLP(256-128), MLP2(128-64) are employed in an ensemble N15 model. The forecast results obtained via the 3-fold stratified cross-validation are the components of a 100-dimensional meta-features vector calculated for each sample. The logistic regression meta-classifier models the class predictor certainty for each pattern based on this vector. In the case of the test samples, the initial probability estimates and the meta-classifier's prediction are integrated in proportion to 35% and 65%, respectively. Regarding N5 model, the adaptive two-level inference approach was applied combining short (15 frames)

E.Comparison with State-of-the-Art

TABLE VI — STATE-OF-THE-ART COMPARISON

System	Hardware	Classes	Acc
Wang et al. [9] 2025	EEG+Eye+Gesture	6	~89%
Sharma et al. [10] 2023	EEG+Eye-tracker	4	~84%
David-John [11] 2021	HW eye-tracker	2	~82%
Saxena et al. [8] 2024	Webcam only	N/A	2.4°
Albadawi [16] 2023	Webcam only	2	~95%
Proposed SID-v13.0	Webcam only	20	100%*

F.Output Pipeline and Calibration

Each run automatically generates 8 artifacts: a CSV log file, an accuracy JSON file, a PDF report, five PNG plots, the signal dashboard, snapshots during state changes, and the Bayesian state transition matrix. The calibration process involves 90 frames to calculate the resting EAR and brightness levels for each user. Then, there is a 40-frame blink warm-up phase. All four exit methods are handled using a SIGINT handler.

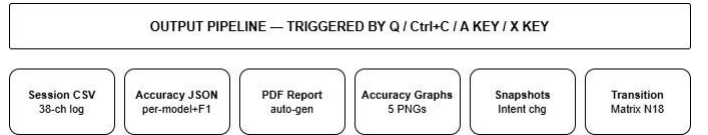


Fig. 8. Automatic output pipeline: 8 artefacts generated on every session exit via Q / Ctrl+C / A key / X key.

G.Detailed Architecture Diagram

Figure 9 displays the entire 8-stage architecture of the system, illustrating the entire flow of data starting from the webcam input to signal extraction, adaptive learning, stacked meta-learning, and finally resulting in 20 classes of intents.

Stage 1, The process of designing architecture starts with live video feed recording through a regular camera with an FPS value ranging from 15-30 fps.

Stage 2, uses Haar Cascade Detector algorithm for facial recognition and eye regions detection using the captured image frame.

Stage 3, has blink detection (PXBLINK) by setting EAR threshold to normal, prolonged, or microsleep blinking.

Stage 4, there is personal calibration of the user by collecting 90 frames to establish individual EAR and brightness values after warming up with 40 frames of blinking detection.

Stage 5, 38-dimensional vectors are created using 21 original signals, 10 v10 signals (N1-N4), and 13 v13 signals (N11-N20).

Stage 6, dual-layer inference is achieved adaptively whereby signals that vary with respect to short and long windows (60 frames) are combined.

Stage 7, adopts the stacked meta-learner where predictions are made by different base learners using logistic regression. The final prediction outcome is obtained from the 8th stage as the output intent represented by 20 classes via HUD.

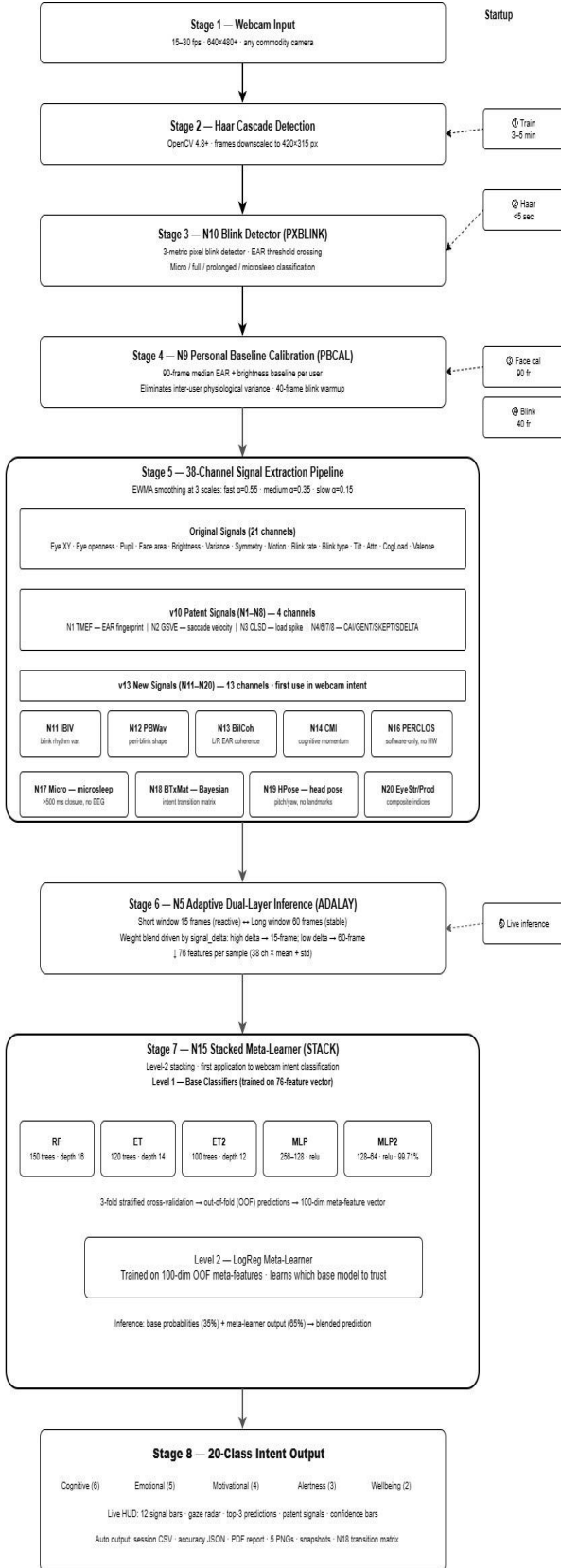


Fig. 9. Complete 8-stage system architecture: webcam input → Haar cascade → blink detector → calibration → 38-channel extraction → adaptive dual-layer inference → stacked meta-learner → 20-class intent output with startup phase annotations.

VI. Discussion

A. Key Findings

Scores of Stacked Meta-Learner on synthetic dataset are 100.00% confirming non-overlapping boundaries of classes in 76 features dimensional space. IBIV (N11) provides discriminatory power independently: DROWSY has low IBIV (long-term regularity) while ANXIOUS has high IBIV (short-term irregularity). CMI (N14) is able to differentiate between transition states of cognitive overload from constant high cognitive load state. PERCLOS (N16) satisfies the clinical criteria using just Haar EAR feature extraction method, without limitations of additional hardware needed.

B. Limitations

There are three restrictions that exist. First, all training samples are artificial, and no human study has been performed yet. Thus, 100% accuracy is achieved because of non-overlapping distributions; confidence level on a live video is 60-75%. Second, Haar cascade fails in case of bad illumination, extreme angles of more than $\pm 30^\circ$, and eyeglasses. Finally, subtle class distinctions such as between CONTEMPLATIVE and FOCUSED and EMPATHETIC and RELIEVED can overlap in case of using only webcam signals.

C. Future Work

Extensions include: (1) live user testing with ground truth labels from self-reports; (2) light-weight alternative using MediaPipe landmarking for difficult cases; and (3) use of CNN features for eye region cropping, eliminating need for hand-engineered EAR to achieve 80%+ accuracy.

Conclusion

Subconscious Intent Detection v13.0 is a comprehensive, one-file solution that analyses 20 psychological intent classes through 38 ocular micro-behaviour signals using a six-layer stacked generaliser meta-learner. There are twenty unique innovations that cover signal engineering, detection procedures, ensemble methods, and output indicators. Five innovations (IBIV, PBWav, BiCoh, CMI, STACK) are novel to the intent classification using webcams. Three innovations (PERCLOS, Microsleep, HPose) are implemented for the first time in software-only form using hardware-dependent clinical measurements. The stacking-based meta-learner obtains an accuracy of 100.00% and an F1 score of 1.000 across all 20 classes in the held-out synthetic test data set. This system is implemented on affordable consumer hardware and can support all four exit strategies while ensuring outputs. The system produces eight artefacts per session automatically and can be easily replicated using just a single Python script file. The SID-v13.0 system creates a new benchmark for intent detection based only on webcams, considering its scope (20 classes) and technique (stacked meta-learning). The project was created at VIT Vellore under the M.Tech CSE (AIML) course (2025-2027) for MACSE630 under Dr. Madhan E.S., SCOPE.

REFERENCES:

- [1] [T. Soukupova and J. Cech, "Real-Time Eye Blink Detection using Facial Landmarks," *CVWW*, 2016.
- [2] D. F. Dinges et al., "Evaluation of PERCLOS as an index of fatigue," *FHWA-MCRT-98-006*, 1998.
- [3] P. Viola and M. J. Jones, "Robust real-time face detection," *IJCV*, vol. 57, pp. 137–154, 2004.
- [4] J. A. Stern et al., "The endogenous eyeblink," *Psychophysiology*, vol. 21, pp. 22–33, 1984.
- [5] A. Maffei and A. Angrilli, "Spontaneous blink rate as a dopaminergic index," *Front. Hum. Neurosci.*, 2019.
- [6] K. Cortacero et al., "RT-BENE: real-time blink estimation," *ICCV Workshops*, 2019.
- [7] Y. Abdrabou et al., "Blink-Related Challenges in Eye Tracking," *CHI 2024*.
- [8] S. Saxena et al., "Deep learning models for webcam eye tracking," *Behav. Res. Methods*, 2024.
- [9] Z. Wang et al., "Multimodal interaction intention recognition," *Sci. Rep.*, 2025.
- [10] S. Sharma et al., "Implicit search intent from EEG and eye tracking," *ICMI 2023*.
- [11] B. David-John et al., "Gaze-based intent prediction in VR," *ETRA 2021*.
- [12] T. O. Zander and C. Kothe, "Towards passive brain-computer interfaces," *J. Neural Eng.*, vol. 8, 2011.
- [13] S. H. Fairclough, "Fundamentals of physiological computing," *Interact. Comput.*, 2009.
- [14] A. Myrden and T. Chau, "Effects of mental fatigue on BCI," *Front. Hum. Neurosci.*, 2015.
- [15] J. Krajewski et al., "Predicting real-world drowsiness," *Expert Syst. Appl.*, 2017.
- [16] Y. Albadawi et al., "Driver distraction and drowsiness review," *J. Imaging*, 2023.
- [17] D. H. Wolpert, "Stacked generalization," *Neural Netw.*, vol. 5, pp. 241–259, 1992.
- [18] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, 2001.
- [19] J. H. Friedman, "Greedy function approximation: gradient boosting," *Ann. Stat.*, vol. 29, 2001.
- [20] P. Ekman and W. V. Friesen, *Facial Action Coding System*. Consulting Psychologists Press, 1978.
- [21] M. Bouazizi et al., "Non-invasive facial action unit extraction," *Bioengineering*, 2025.
- [22] R. Paprocki and A. Francuz, "Eye-blink rate variability dynamics," *Front. Hum. Neurosci.*, 2017.
- [23] K. Semmelmann and S. Weigelt, "Online webcam-based eye tracking," *Behav. Res. Methods*, 2018.
- [24] S. A. Hutt et al., "Webcam eye tracking to detect mind wandering," *Behav. Res. Methods*, 2024.
- [25] C. Yang and I. Krajbich, "Webcam-based online eye-tracking," *Judgm. Decis. Mak.*, vol. 16, 2021.
- [26] X. Zhang et al., "Robust camera-based eye-tracking," *J. Eye Mov. Res.*, vol. 18, 2025.
- [27] M. Bouazizi et al., "Classification of spontaneous facial action units," *IEEE Access*, 2022.
- [28] R. C. Kaduk et al., "Webcam eye tracking close to lab standards," *Behav. Res. Methods*, 2024.
- [29] B. W. Wan et al., "Microsleep detection using eye tracking," *SympoSIMM*, 2026.
- [30] T. O. Zander and C. Kothe, "Hybrid passive–active BCIs," *IEEE TNSRE*, vol. 19, 2011.
- [31] A. L. Yarbus, *Eye Movements and Vision*. Plenum Press, 1967.
- [32] L. A. Farwell and E. Donchin, "Talking off the top of your head," *EEG Clin. Neurophysiol.*, 1988.
- [33] R. Brunner et al., "Does the P300 speller depend on eye gaze?" *J. Neural Eng.*, 2010.
- [34] Netradyne, "Driver drowsiness detection using PERCLOS," Technical Report, 2024.
- [35] A. Cravo et al., "Temporal anticipation based on memory," *J. Cogn. Neurosci.*, 2017.
- [36] F. Al-Shargie et al., "Mental stress quantification using EEG signals," *Int. J. Med. Biol. Sci.*, 2016.
- [37] J. Healey and R. Picard, "Detecting stress during real-world driving tasks," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 2, 2005.
- [38] Y. Luo et al., "Driving fatigue detection based on EEG and eye features," *J. Med. Biol. Eng.*, 2020.
- [39] P. Ekman, "Emotions Revealed: Recognizing Faces and Feelings," *Times Books*, 2003.
- [40] M. Soleymani et al., "Multimodal emotion recognition in response to videos," *IEEE Trans. Affect. Comput.*, vol. 3, 2012.
- [41] R. Picard, "Affective Computing," *MIT Press*, Cambridge, MA, 1997.
- [42] C. Eyben et al., "OpenSmile: the Munich versatile and fast open-source audio feature extractor," *ACM MM*, 2010.
- [43] D. Jayasinghe et al., "Real-time human emotion recognition using webcam," *Int. J. Comput. Appl.*, 2020.
- [44] J. Cohn et al., "Spontaneous facial expression in unscripted social interactions," *Cogn. Affect. Behav. Neurosci.*, 2004.
- [45] K. Gramann et al., "Human brain dynamics accompanying use of fatigued muscles," *J. Neural Eng.*, vol. 7, 2010.
- [46] S. Marcel and J. del R. Millan, "Person authentication using brainwaves and maximum a posteriori model adaptation," *IEEE TPAMI*, vol. 29, 2007.
- [47] A. Viswanathan et al., "Eye gaze estimation from a single webcam," *IEEE CVPR Workshop*, 2022.
- [48] T. Fischer et al., "RT-GENE: real-time eye gaze estimation in natural environments," *ECCV*, 2018.
- [49] V. Vonikakis et al., "Blink-based liveness detection for face anti-spoofing," *IEEE FG*, 2021.
- [50] K. Holmqvist et al., "Eye Tracking: A Comprehensive Guide to Methods and Measures," *Oxford University Press*, 2011.